HELLO I AM...

MOHAMMAD AARISH KHAN

BIOMEDICAL ENGINEERING MAJOR

University at Buffalo
Honors College
August 2026 - Present

Williamsville North High School
Class of 2026

WORKSHOPS

- MIT Beaver Works Summer Institute — Autonomous Cars Course
- Biotech in Action — Biogen Community Lab & Lemelson-MIT
- Oracle Cloud Infrastructure Gen AI Certification
- Eric Pitman Summer Workshop — Intern Computational Biologist

CONFERENCES

- Kennedy Space Center 2025 (NASA SmallSat)
- IFoRE 2025
- ACS NERD 2025
- Terra Science Fair
- Global Impact Council Conference
- IEEE ISEC 2025

AWARDS & HONORS

NYAS Emerging Student Researcher of the Year (2025):

- Top 5 of ~8,000 internationally
- 1st Place (2023)
- Project of Distinction (2024)

ACS NERD Outstanding Poster Presentation Award (2025)

Terra Science Fair (2025):

- Medal of High Honors
- Office of Naval Research Award

President's Volunteer Service Award (2025)
— Bronze
600+ Volunteer hours

CONTACT

kmohammadaarish@gmail.com

aarishkhan.vercel.app

RESEARCH EXPERIENCE

Current Research

Williams Lab

Neurogenetic Research

- Analyze GWAS data sets using FLAMES and FUMA-based analysis to investigate the genetic basis of neurological conditions

Nushoor

Water Purification Project

- Collaborate with a team designing a low-cost, UV-based water purification system fine-tuned for disenfranchised communities
- Tailor system design to community-specific constraints, including limited resources, cost, maintenance, and environmental conditions

Previous Research

First Author Publication

Computational Design for Bispecific Antibodies

2024 - 2025

- Designed a Bispecific DuoBody antibody targeting 4-1BB on cancer cells and PD-L1 on T-cells as a potential treatment for non-small cell lung cancer (NSCLC)
- (DOI: 10.1109/ISEC64801.2025.11147433)
- Built a Streamlit-based web app that automated parts of the workflow

New York Academy of Sciences, Junior Academy

Research Projects

2023-2024

- Head of Ideation and Implementation for two international teams: an asteroid mining project (1st Place) and a multiple sclerosis health wearable (Project of Distinction)
- Co-authored abstracts and PowerPoint presentations for both projects
- Selected for the Emerging Student Researcher of the Year award

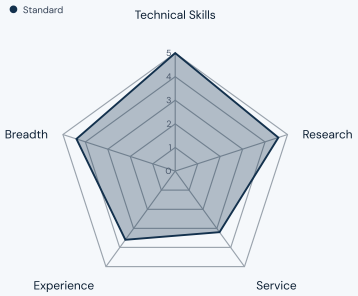
University at Buffalo Nanosatellite Lab

Kennedy Space Center Presentation

2025

- Designed electronics and programming as part of a 5-student team for a high-altitude balloon mission carrying six solar panels
- Studied variation in solar panel efficiency across altitude
- Presented findings at the SmallSat Conference, Kennedy Space Center (NASA)

STATS



6 Presentations

1 Paper

5 Abstracts

4+ Labs

LEADERSHIP

Biomedical Engineering Society | 2026 - Present
Battle Bots Team Lead